

Practical Assignment:

Router Configuration and **Routing Protocols using** **Cisco Packet**

Module:

Computer Networks

Task 1 – Basic Router Configuration

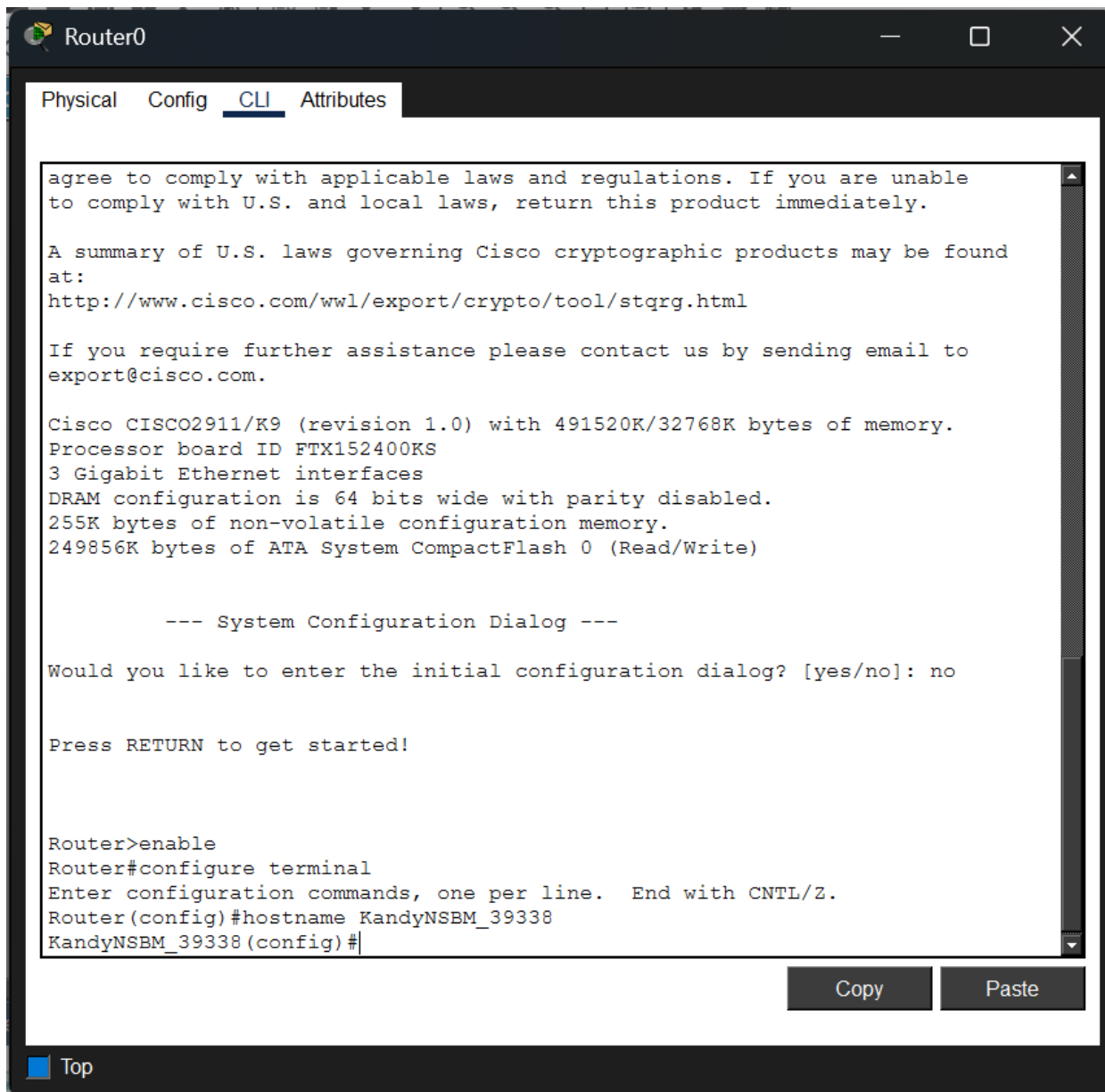


Figure 1 hostname

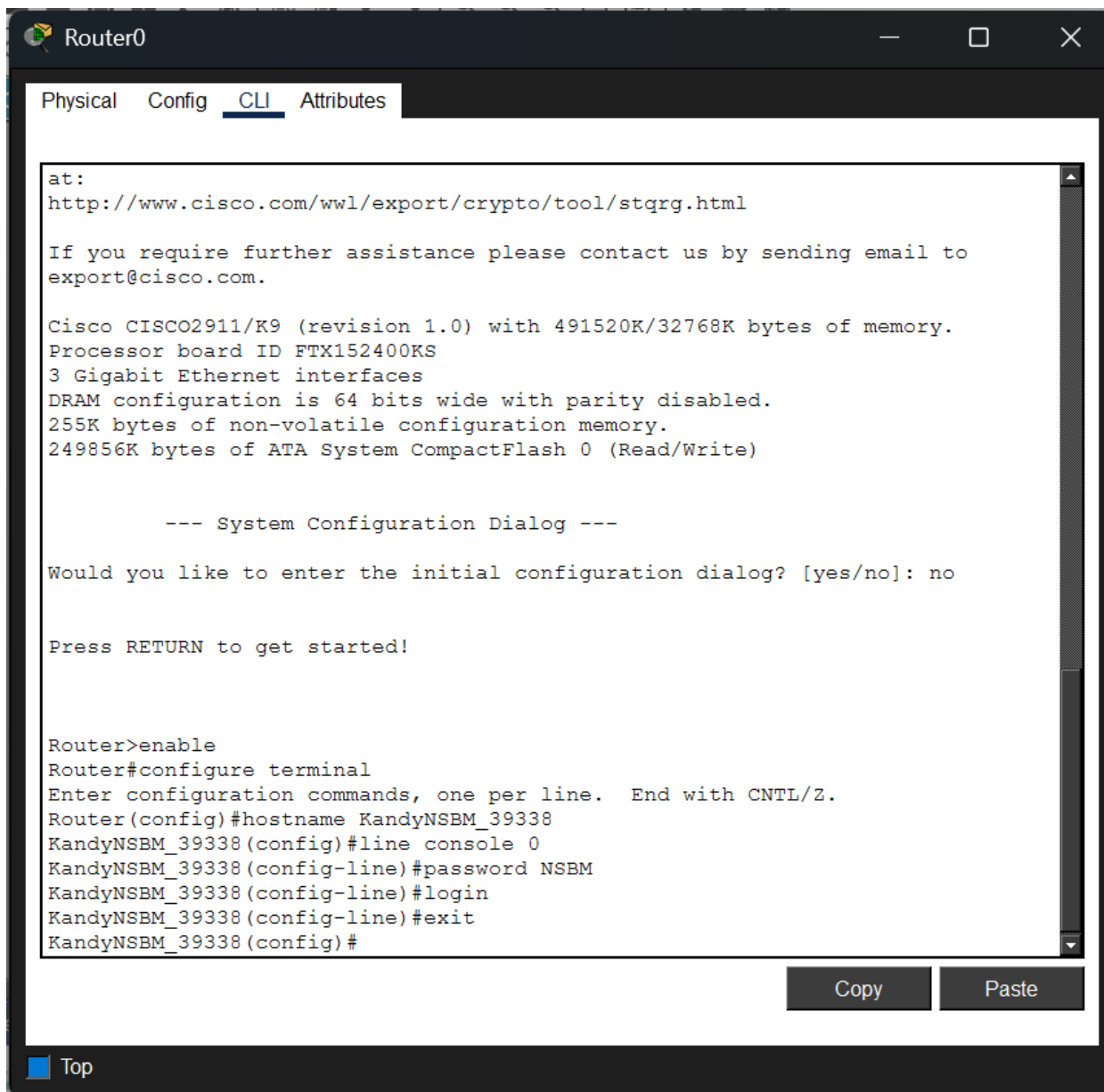
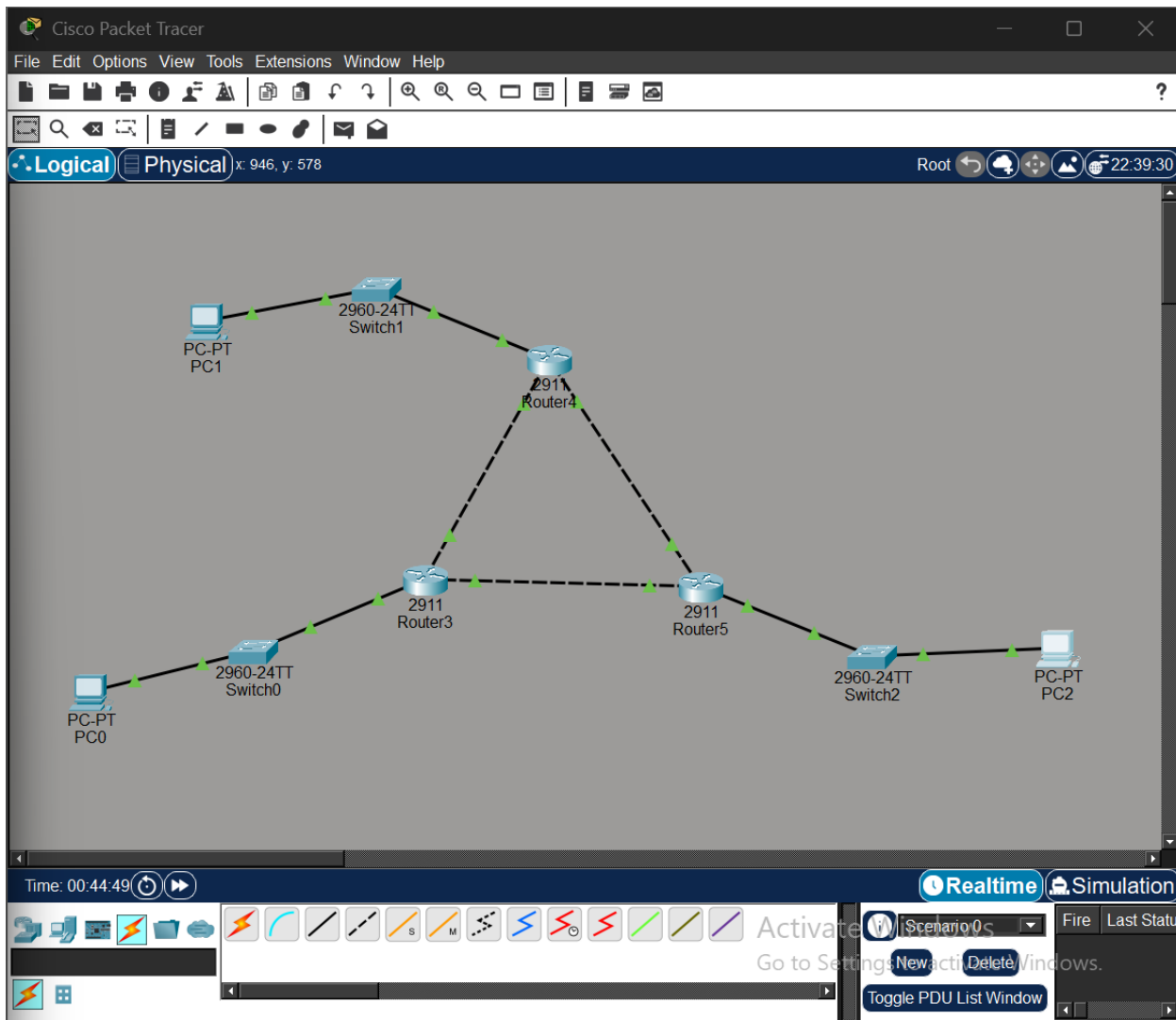


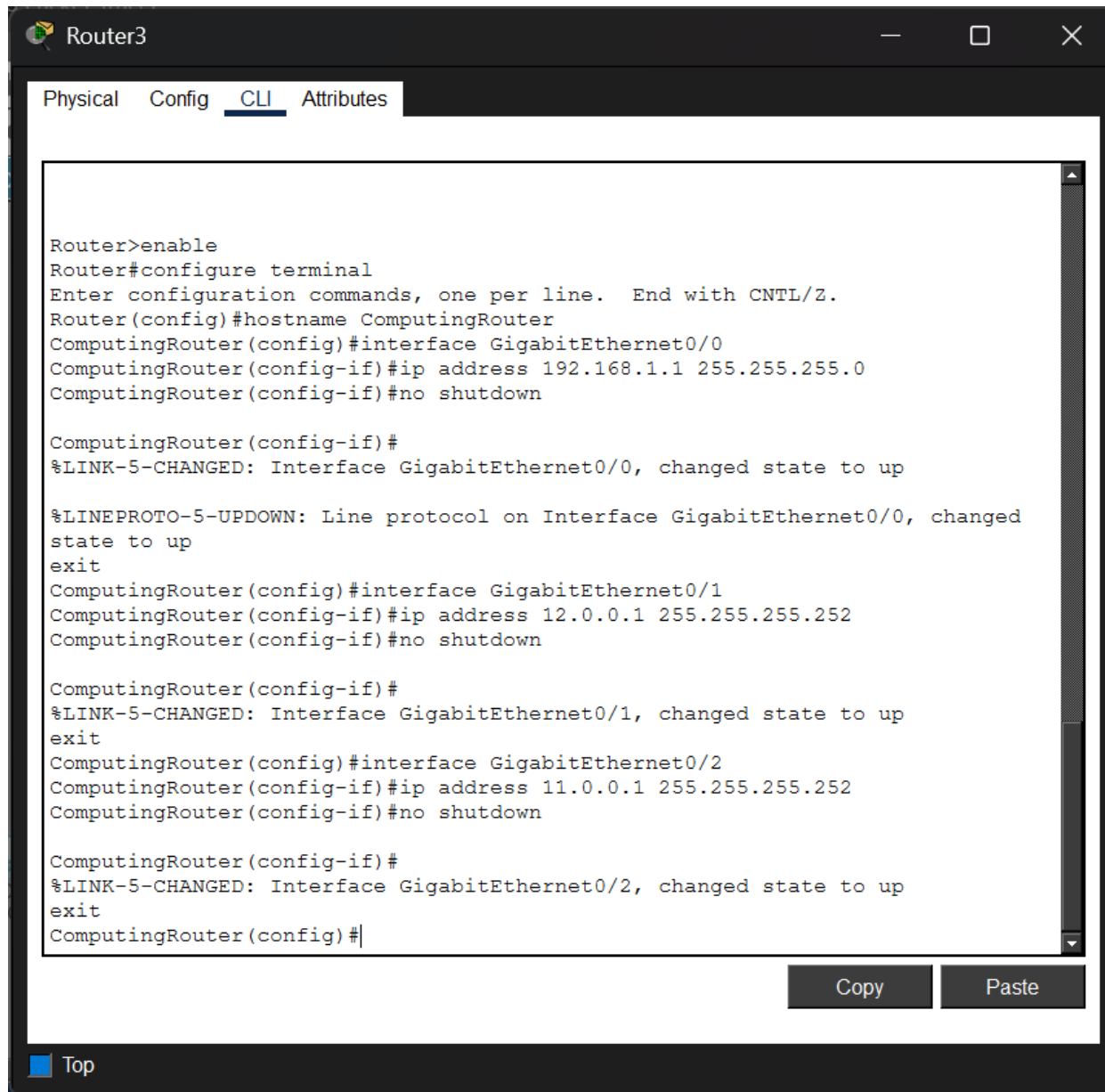
Figure 2 Password

Task 2 – Static Routing

1. Network diagram with IP addresses



2. Configuration commands used on each router



The screenshot shows a Cisco Packet Tracer window titled "Router3". It has four tabs: "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is selected. The terminal window displays the following commands and output:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ComputingRouter
ComputingRouter(config)#interface GigabitEthernet0/0
ComputingRouter(config-if)#ip address 192.168.1.1 255.255.255.0
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up
exit
ComputingRouter(config)#interface GigabitEthernet0/1
ComputingRouter(config-if)#ip address 12.0.0.1 255.255.255.252
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
ComputingRouter(config)#interface GigabitEthernet0/2
ComputingRouter(config-if)#ip address 11.0.0.1 255.255.255.252
ComputingRouter(config-if)#no shutdown

ComputingRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
exit
ComputingRouter(config)#
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. At the bottom left of the window, there is a "Top" button.

```
%SYS-5-CONFIG_I: Configured from console by console
end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

BuisnessRouter#
BuisnessRouter#Interface GigabitEthernet0/0
      ^
% Invalid input detected at '^' marker.

BuisnessRouter#Interface GigabitEthernet0/2
      ^
% Invalid input detected at '^' marker.

BuisnessRouter#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
BuisnessRouter(config)#interface GigabitEthernet0/0
BuisnessRouter(config-if)#ip address 192.168.2.1 255.255.255.0
BuisnessRouter(config-if)#no shutdown
BuisnessRouter(config-if)#exit
BuisnessRouter(config)#interface GigabitEthernet0/1
BuisnessRouter(config-if)#ip address 12.0.0.2 255.255.255.252
BuisnessRouter(config-if)#no shutdown
BuisnessRouter(config-if)#exit
BuisnessRouter(config)#interface GigabitEthernet0/2
BuisnessRouter(config-if)#ip address 10.0.0.2 255.255.255.252
BuisnessRouter(config-if)#no shutdown

BuisnessRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
exit
BuisnessRouter(config)#
```

Copy

Paste

```
Router(config)#hostname ScienceRouter
ScienceRouter(config)#interface GigabitEthernet0/0
ScienceRouter(config-if)#ip address 192.168.3.1 255.255.255.0
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up
exit
ScienceRouter(config)#interface GigabitEthernet0/1
ScienceRouter(config-if)#ip address 11.0.0.2 255.255.255.252
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up
exit
ScienceRouter(config)#interface GigabitEthernet0/2
ScienceRouter(config-if)#ip address 10.0.0.1 255.255.255.252
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed
state to up
exit
ScienceRouter(config)#
```

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Press RETURN to get started.

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

ComputingRouter>ip route 192.168.2.0 255.255.255.0 12.0.0.2

^
% Invalid input detected at '^' marker.

ComputingRouter>configure terminal

^
% Invalid input detected at '^' marker.

ComputingRouter>enable

ComputingRouter#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

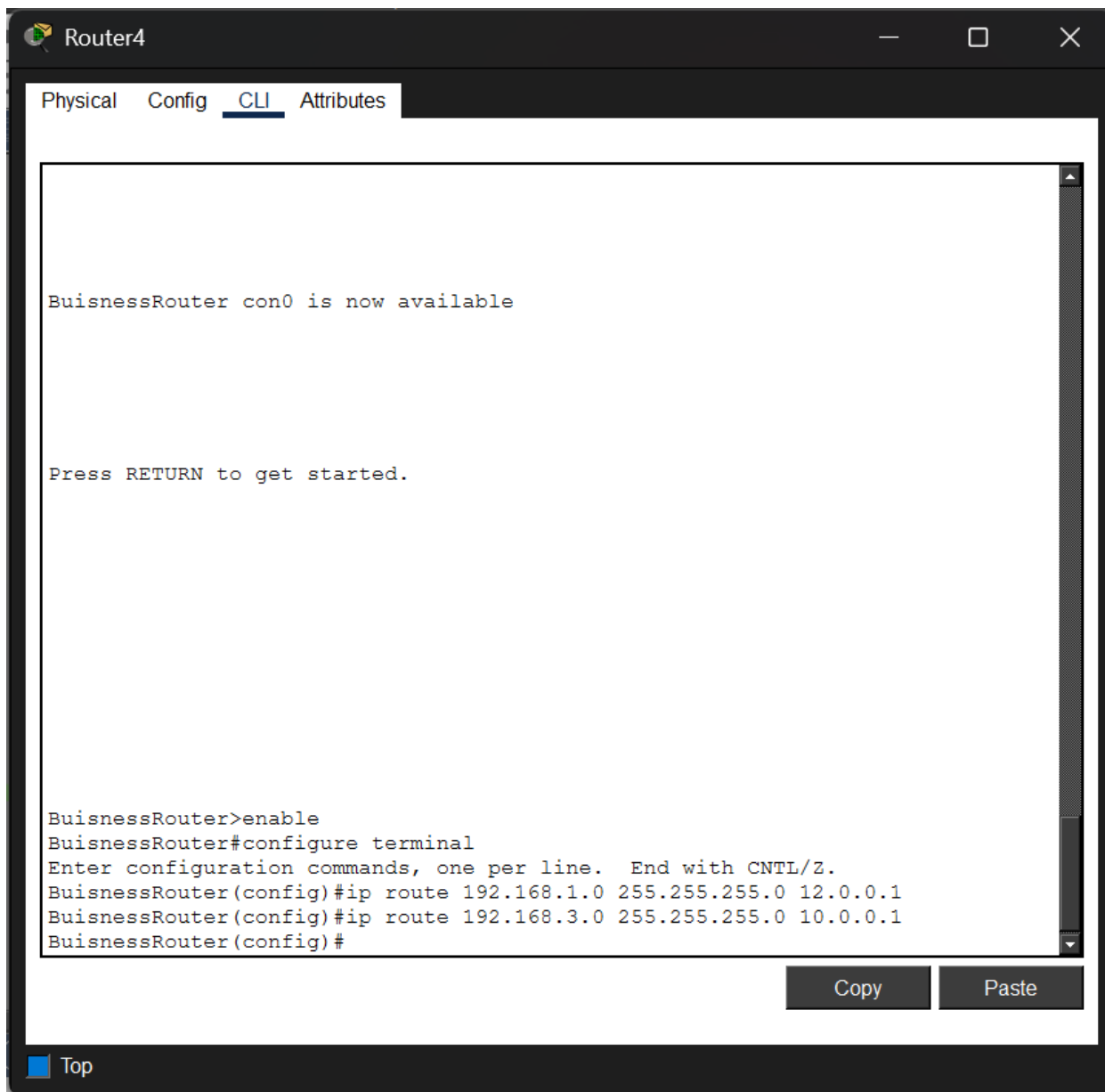
ComputingRouter(config)#ip route 192.168.2.0 255.255.255.0 12.0.0.2

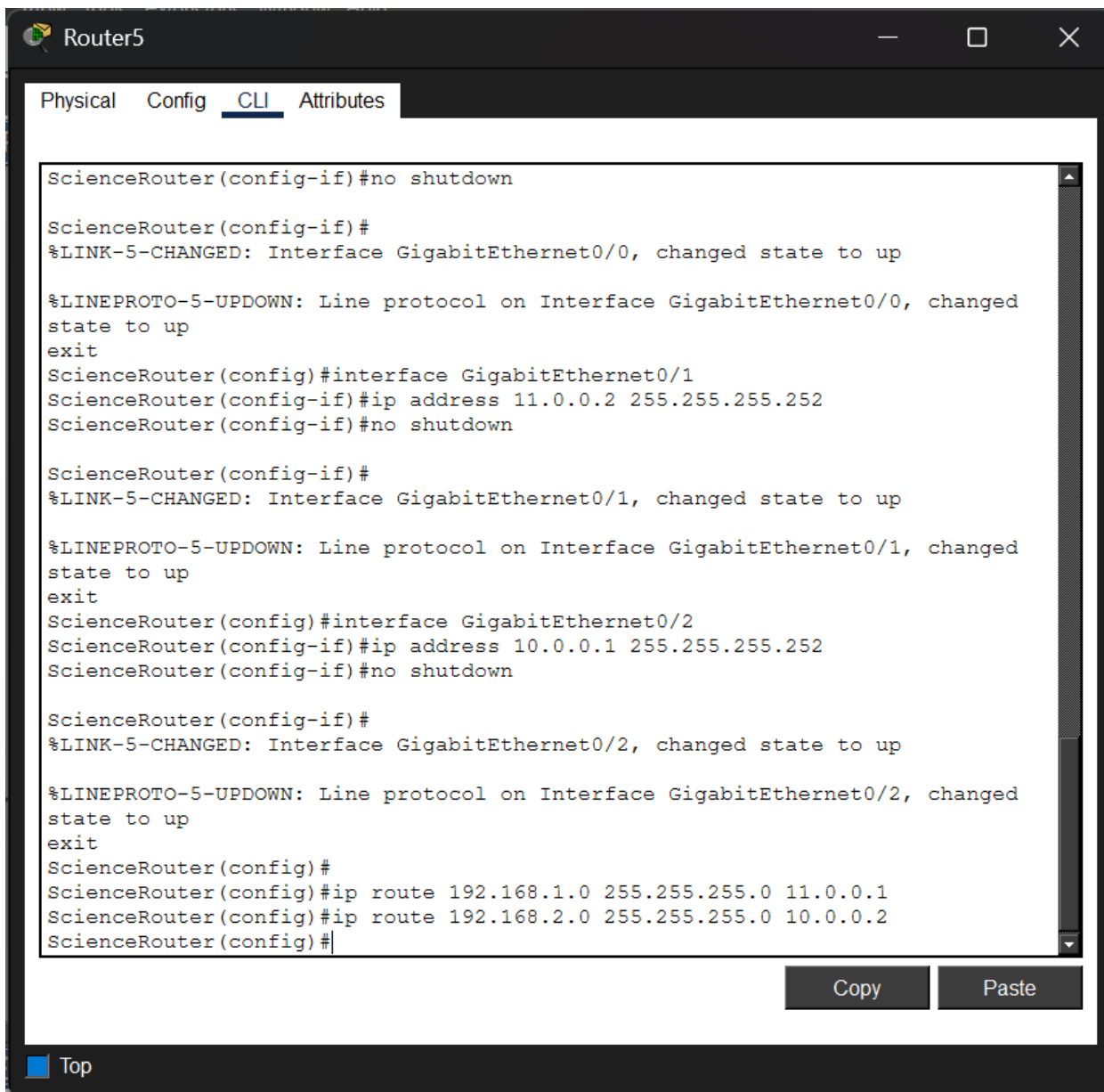
ComputingRouter(config)#ip route 192.168.3.0 255.255.255.0 11.0.0.2

ComputingRouter(config)#

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```
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up
exit
ScienceRouter(config)#interface GigabitEthernet0/1
ScienceRouter(config-if)#ip address 11.0.0.2 255.255.255.252
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed
state to up
exit
ScienceRouter(config)#interface GigabitEthernet0/2
ScienceRouter(config-if)#ip address 10.0.0.1 255.255.255.252
ScienceRouter(config-if)#no shutdown

ScienceRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed
state to up
exit
ScienceRouter(config)#
ScienceRouter(config)#ip route 192.168.1.0 255.255.255.0 11.0.0.1
ScienceRouter(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
ScienceRouter(config)#
```

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Top

3. Routing table screenshots

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3. Routing table screenshots

```
Router3
Physical Config CLI Attributes
ComputingRouter(config)#show ip route
^
% Invalid input detected at '^' marker.
ComputingRouter(config)#exit
ComputingRouter#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, GigabitEthernet0/2
L       11.0.0.1/32 is directly connected, GigabitEthernet0/2
    12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/30 is directly connected, GigabitEthernet0/1
L       12.0.0.1/32 is directly connected, GigabitEthernet0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S       192.168.2.0/24 [1/0] via 12.0.0.2
S       192.168.3.0/24 [1/0] via 11.0.0.2

ComputingRouter#
```

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Physical Config CLI Attributes

```
BuisnessRouter#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
BuisnessRouter(config)#ip route 192.168.1.0 255.255.255.0 12.0.0.1
BuisnessRouter(config)#ip route 192.168.3.0 255.255.255.0 10.0.0.1
BuisnessRouter(config)#exit
BuisnessRouter#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.2/32 is directly connected, GigabitEthernet0/2
  12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       12.0.0.0/30 is directly connected, GigabitEthernet0/1
L       12.0.0.2/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 12.0.0.1
       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0
S       192.168.3.0/24 [1/0] via 10.0.0.1

BuisnessRouter#
```

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```
exit
ScienceRouter(config)#
ScienceRouter(config)#ip route 192.168.1.0 255.255.255.0 11.0.0.1
ScienceRouter(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
ScienceRouter(config)#exit
ScienceRouter#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

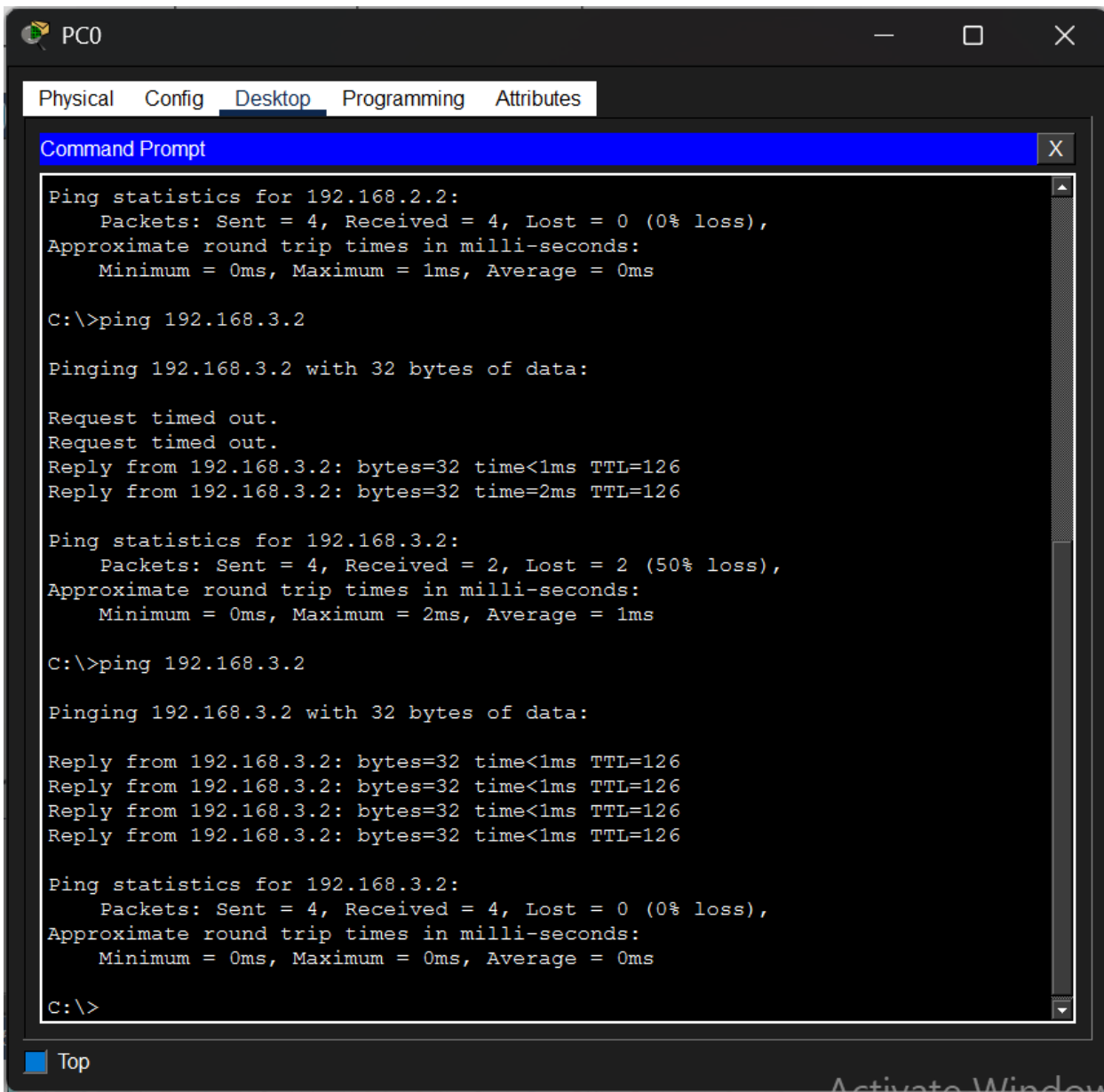
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/2
L       10.0.0.1/32 is directly connected, GigabitEthernet0/2
    11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, GigabitEthernet0/1
L       11.0.0.2/32 is directly connected, GigabitEthernet0/1
S       192.168.1.0/24 [1/0] via 11.0.0.1
S       192.168.2.0/24 [1/0] via 10.0.0.2
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0

ScienceRouter#
```

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4. Ping test screenshots



The screenshot shows a window titled "PC0" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of two ping tests. The first test is for 192.168.2.2, which shows 0% loss and 0ms average round trip time. The second test is for 192.168.3.2, which shows 50% loss and 1ms average round trip time. The Command Prompt window has a blue title bar and a scroll bar on the right.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 192.168.3.2

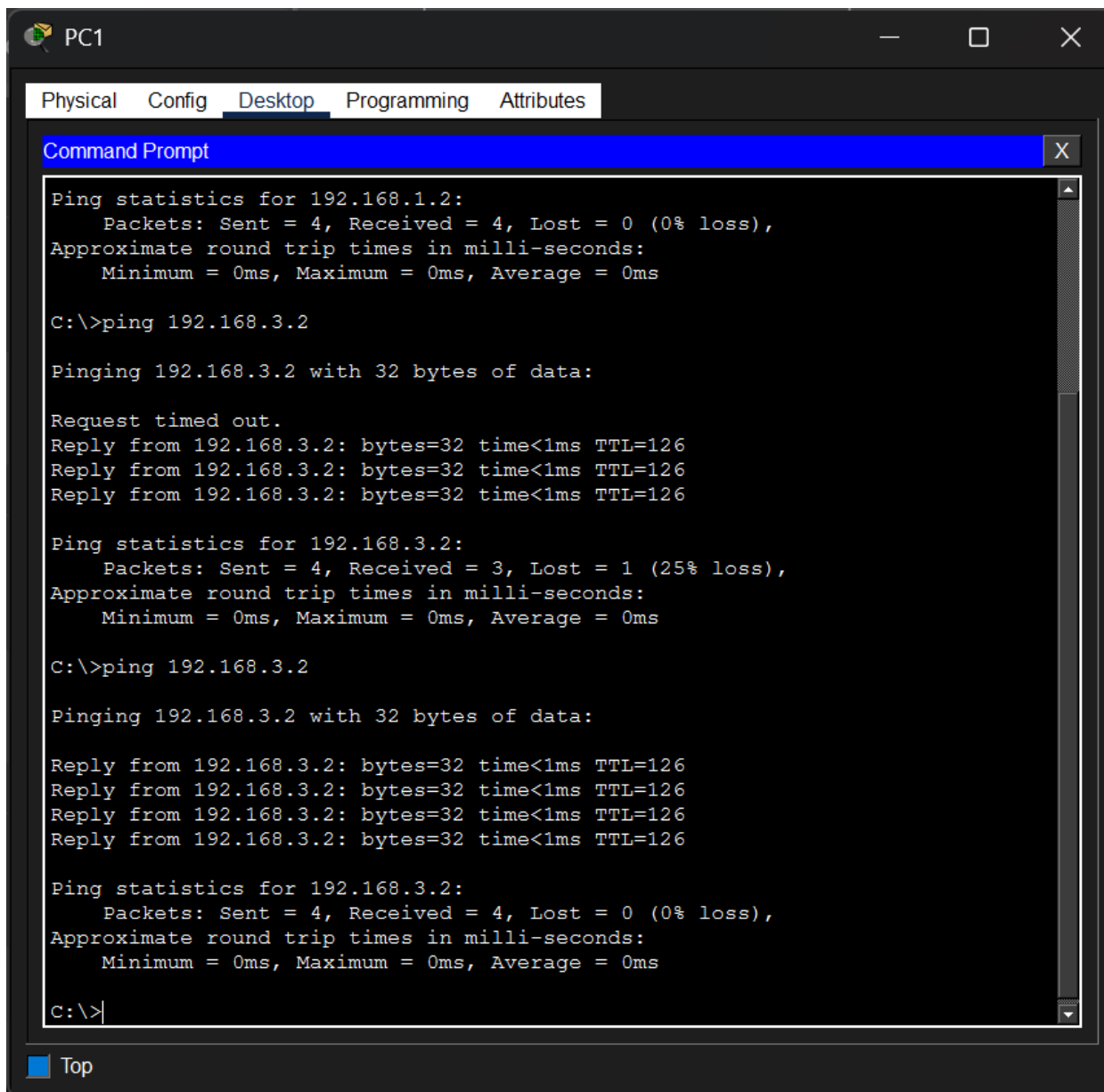
Pinging 192.168.3.2 with 32 bytes of data:

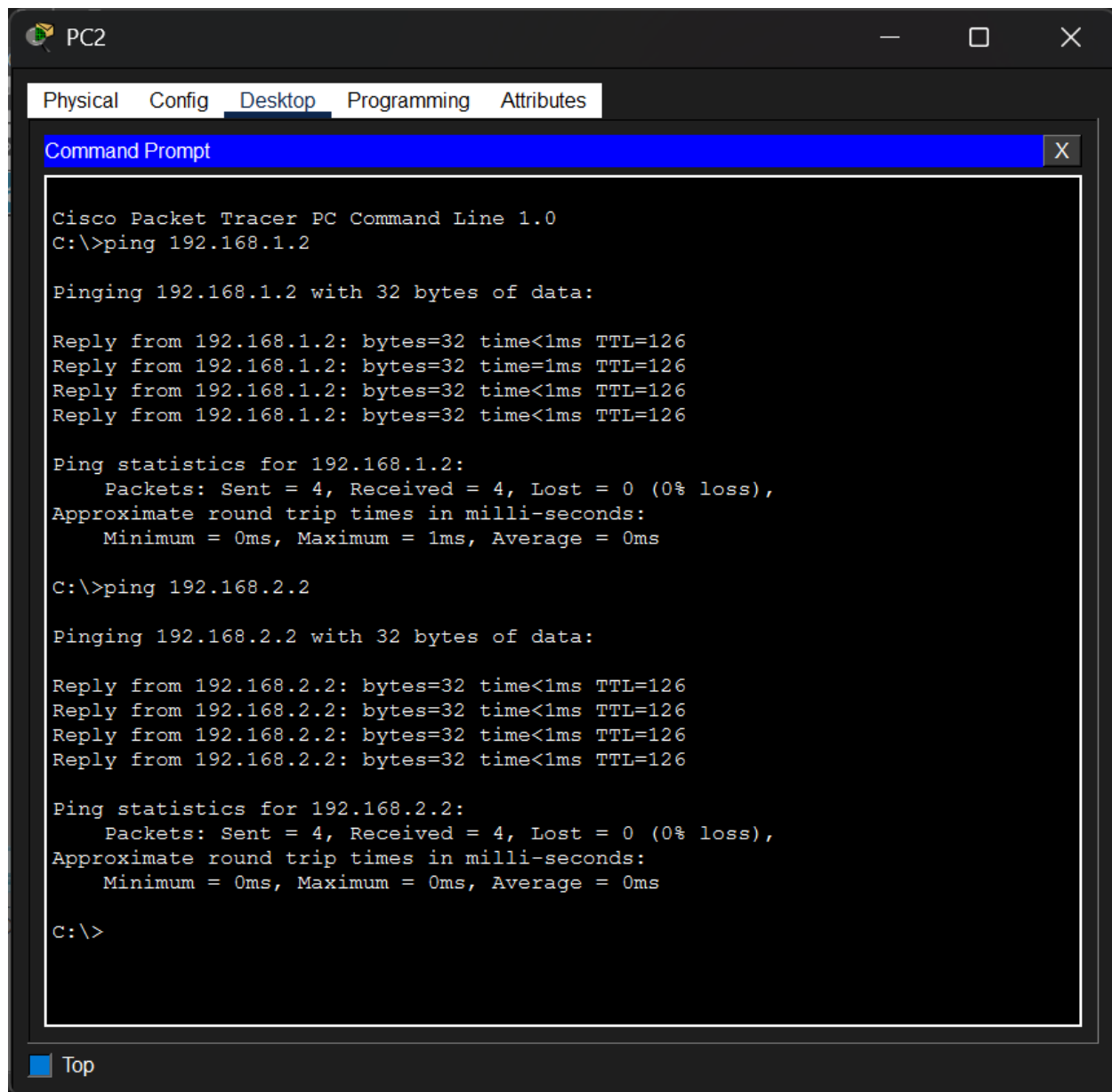
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126
Reply from 192.168.3.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Top





The screenshot shows a window titled "PC2" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the output of two ping commands: "ping 192.168.1.2" and "ping 192.168.2.2". Both commands show successful results with 0% loss and 0ms average round trip time. The Command Prompt window has a blue title bar and a close button (X) in the top right corner. At the bottom left of the PC2 window, there is a "Top" button.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

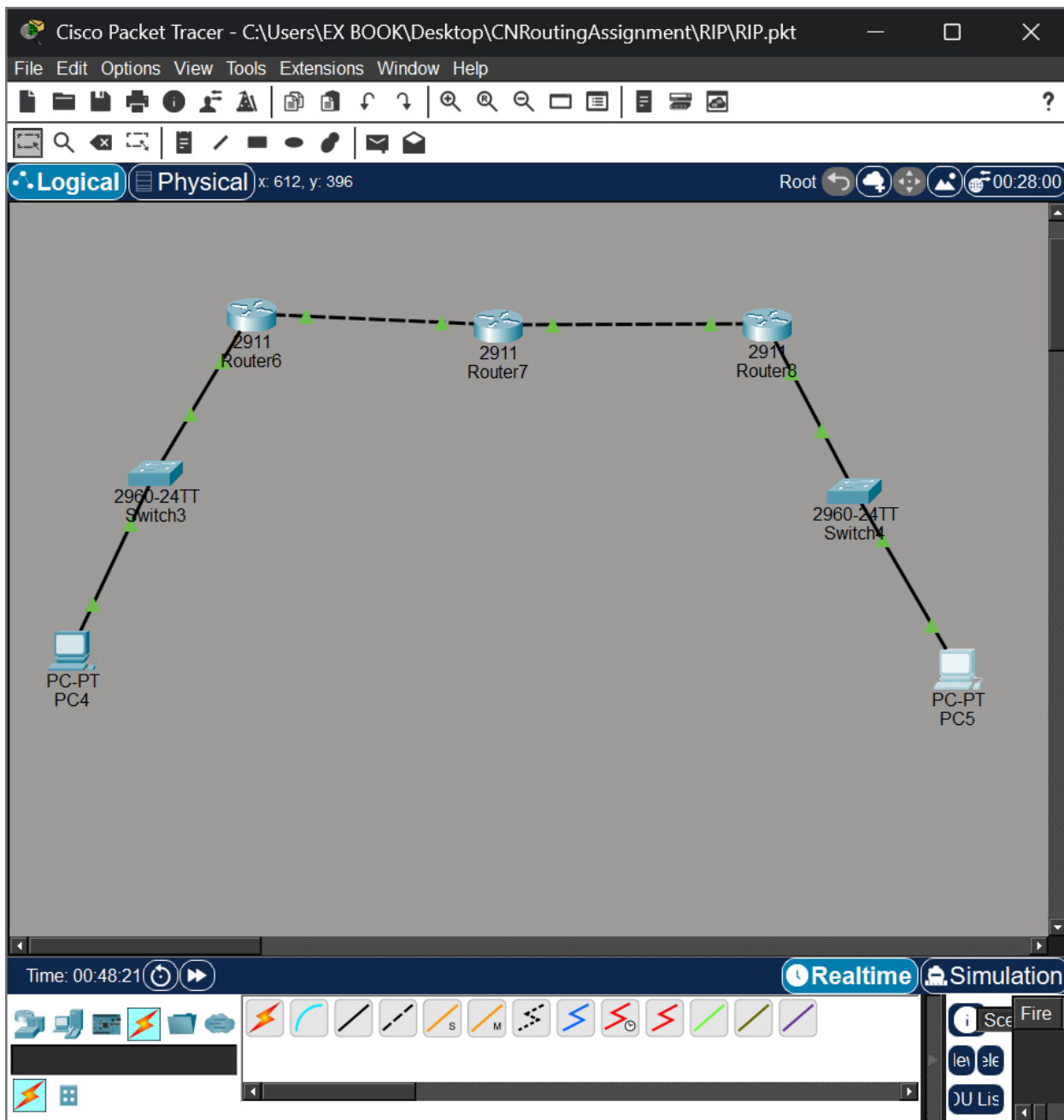
Brief Explanation on Static Routing Configuration

In this task, three routers and three separate networks were configured using IP addressing and static routing. Static routes were manually added on each router using the ip route command to allow communication between different networks. Connectivity between all networks was verified successfully using ping tests. Static routing provides fixed paths for data transmission and is suitable for small networks.

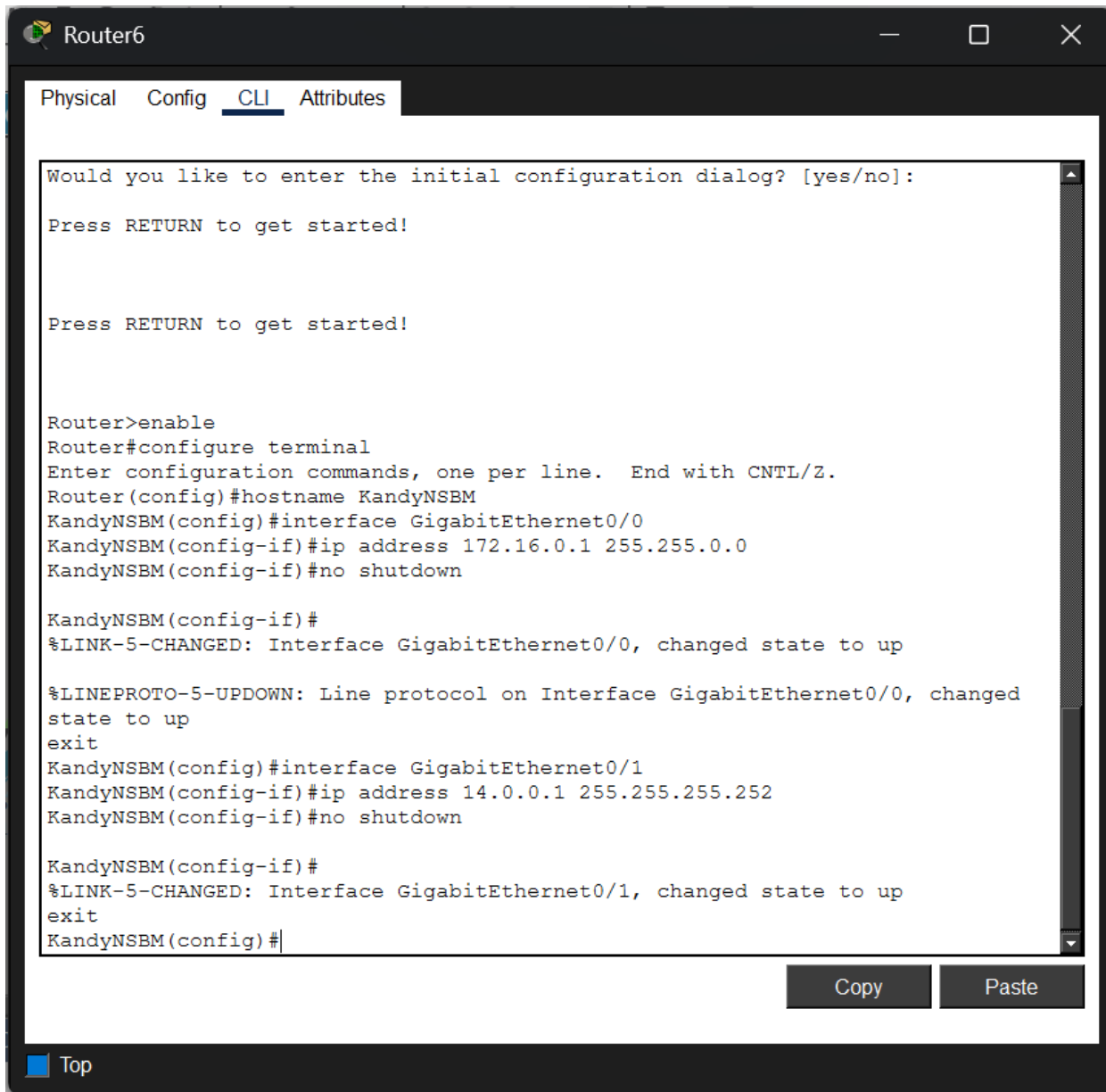
Task 3 – Dynamic Routing Configuration

Part A – RIP Routing

1. Network diagram



2. Configuration commands



The screenshot shows a terminal window titled "Router6" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active. The terminal displays the following text:

```
Would you like to enter the initial configuration dialog? [yes/no]:  
Press RETURN to get started!  
  
Press RETURN to get started!  
  
Router>enable  
Router#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#hostname KandyNSBM  
KandyNSBM(config)#interface GigabitEthernet0/0  
KandyNSBM(config-if)#ip address 172.16.0.1 255.255.0.0  
KandyNSBM(config-if)#no shutdown  
  
KandyNSBM(config-if)#  
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed  
state to up  
exit  
KandyNSBM(config)#interface GigabitEthernet0/1  
KandyNSBM(config-if)#ip address 14.0.0.1 255.255.255.252  
KandyNSBM(config-if)#no shutdown  
  
KandyNSBM(config-if)#  
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up  
exit  
KandyNSBM(config)#|
```

At the bottom right of the terminal area, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Press RETURN to get started!

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname ColomboNSBM

ColomboNSBM(config)#interface GigabitEthernet0/0

ColomboNSBM(config-if)#ip address 14.0.0.2 255.255.255.252

ColomboNSBM(config-if)#no shutdown

ColomboNSBM(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

exit

ColomboNSBM(config)#interface GigabitEthernet0/1

ColomboNSBM(config-if)#ip address 15.0.0.1 255.255.255.252

ColomboNSBM(config-if)#no shutdown

ColomboNSBM(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

exit

ColomboNSBM(config)#

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Press RETURN to get started!

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#hostname GalleNSBM

GalleNSBM(config)#interface GigabitEthernet0/0

GalleNSBM(config-if)#ip address 173.16.0.1 255.255.0.0

GalleNSBM(config-if)#no shutdown

GalleNSBM(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

exit

GalleNSBM(config)#interface GigabitEthernet0/1

GalleNSBM(config-if)#ip address 15.0.0.2 255.255.255.252

GalleNSBM(config-if)#no shutdown

GalleNSBM(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

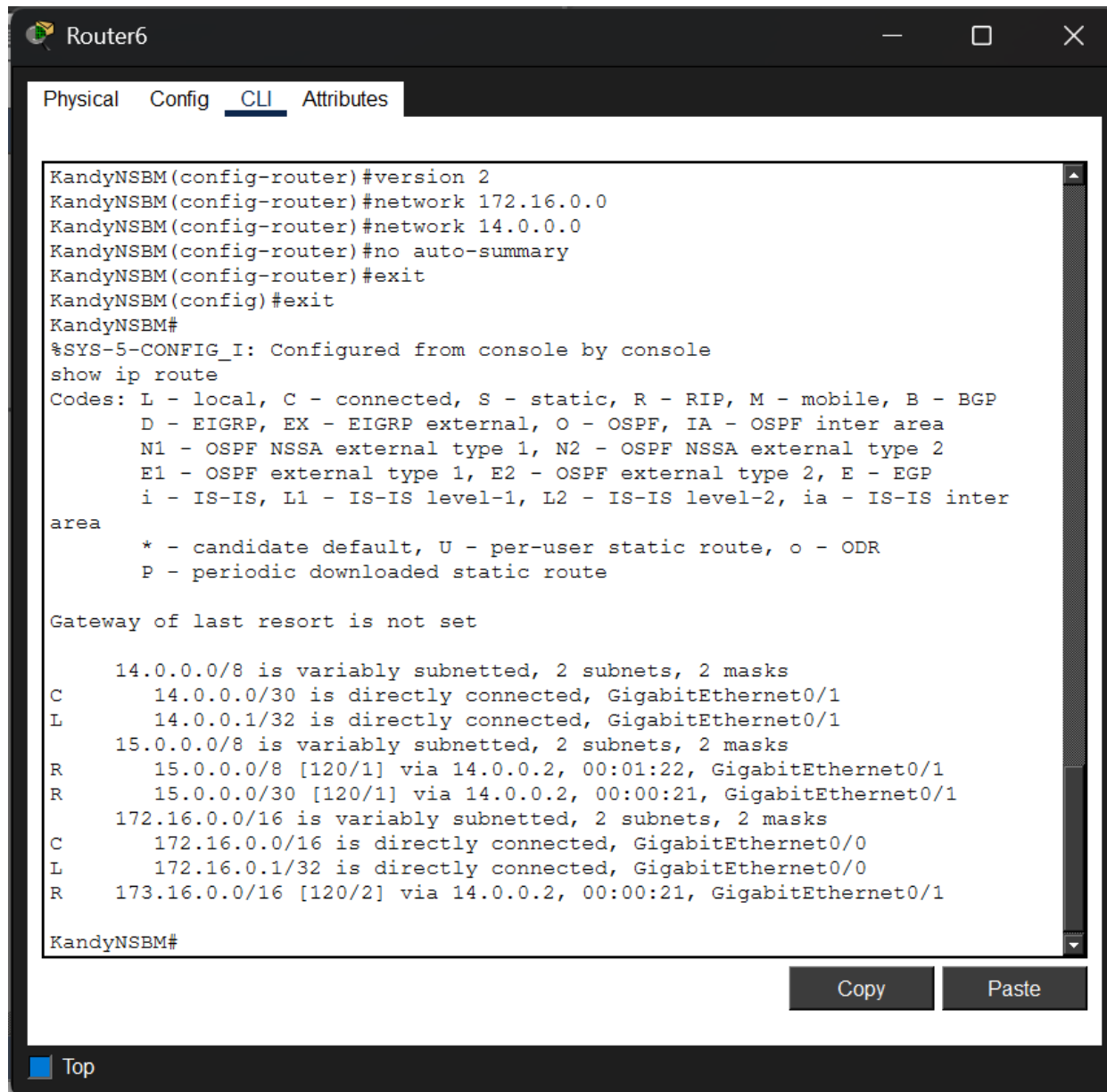
exit

GalleNSBM(config)#

Copy

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4. Screenshots of routing tables



The screenshot shows a terminal window titled "Router6" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the following commands and output:

```
KandyNSBM(config-router)#version 2
KandyNSBM(config-router)#network 172.16.0.0
KandyNSBM(config-router)#network 14.0.0.0
KandyNSBM(config-router)#no auto-summary
KandyNSBM(config-router)#exit
KandyNSBM(config)#exit
KandyNSBM#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/1
L       14.0.0.1/32 is directly connected, GigabitEthernet0/1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
R       15.0.0.0/8 [120/1] via 14.0.0.2, 00:01:22, GigabitEthernet0/1
R       15.0.0.0/30 [120/1] via 14.0.0.2, 00:00:21, GigabitEthernet0/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
R       173.16.0.0/16 [120/2] via 14.0.0.2, 00:00:21, GigabitEthernet0/1

KandyNSBM#
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

```
ColomboNSBM(config)#router rip
ColomboNSBM(config-router)#version 2
ColomboNSBM(config-router)#network 14.0.0.0
ColomboNSBM(config-router)#network 15.0.0.0
ColomboNSBM(config-router)#no auto-summary
ColomboNSBM(config-router)#exit
ColomboNSBM(config)#exit
ColomboNSBM#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/0
L       14.0.0.2/32 is directly connected, GigabitEthernet0/0
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       15.0.0.0/30 is directly connected, GigabitEthernet0/1
L       15.0.0.1/32 is directly connected, GigabitEthernet0/1
R       172.16.0.0/16 [120/1] via 14.0.0.1, 00:00:03, GigabitEthernet0/0
R       173.16.0.0/16 [120/1] via 15.0.0.2, 00:00:05, GigabitEthernet0/1

ColomboNSBM#
```

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```
GalleNSBM(config)#router rip
GalleNSBM(config-router)#version 2
GalleNSBM(config-router)#network 15.0.0.0
GalleNSBM(config-router)#network 173.16.0.0
GalleNSBM(config-router)#no auto-summary
GalleNSBM(config-router)#exit
GalleNSBM(config)#exit
GalleNSBM#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

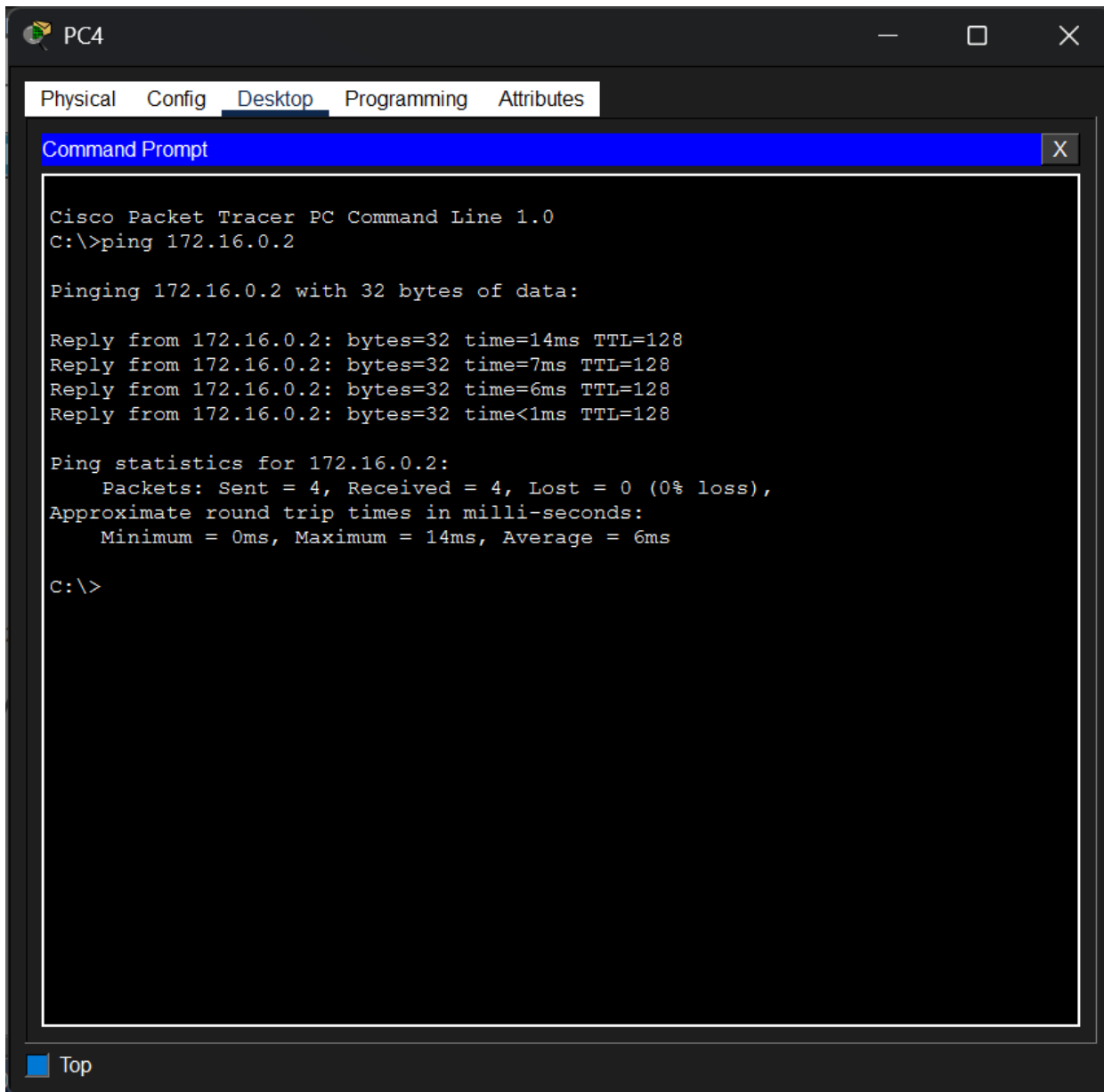
    14.0.0.0/30 is subnetted, 1 subnets
R       14.0.0.0/30 [120/1] via 15.0.0.1, 00:00:02, GigabitEthernet0/1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       15.0.0.0/30 is directly connected, GigabitEthernet0/1
L       15.0.0.2/32 is directly connected, GigabitEthernet0/1
R       172.16.0.0/16 [120/2] via 15.0.0.1, 00:00:02, GigabitEthernet0/1
    173.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       173.16.0.0/16 is directly connected, GigabitEthernet0/0
L       173.16.0.1/32 is directly connected, GigabitEthernet0/0

GalleNSBM#
```

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5. Ping test results



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC4. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, and a Command Prompt window is open. The Command Prompt displays the output of a ping command to 172.16.0.2. The output shows four successful replies with varying times and a TTL of 128. Ping statistics indicate 0% loss and an average round trip time of 6ms.

```
PC4
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.0.2

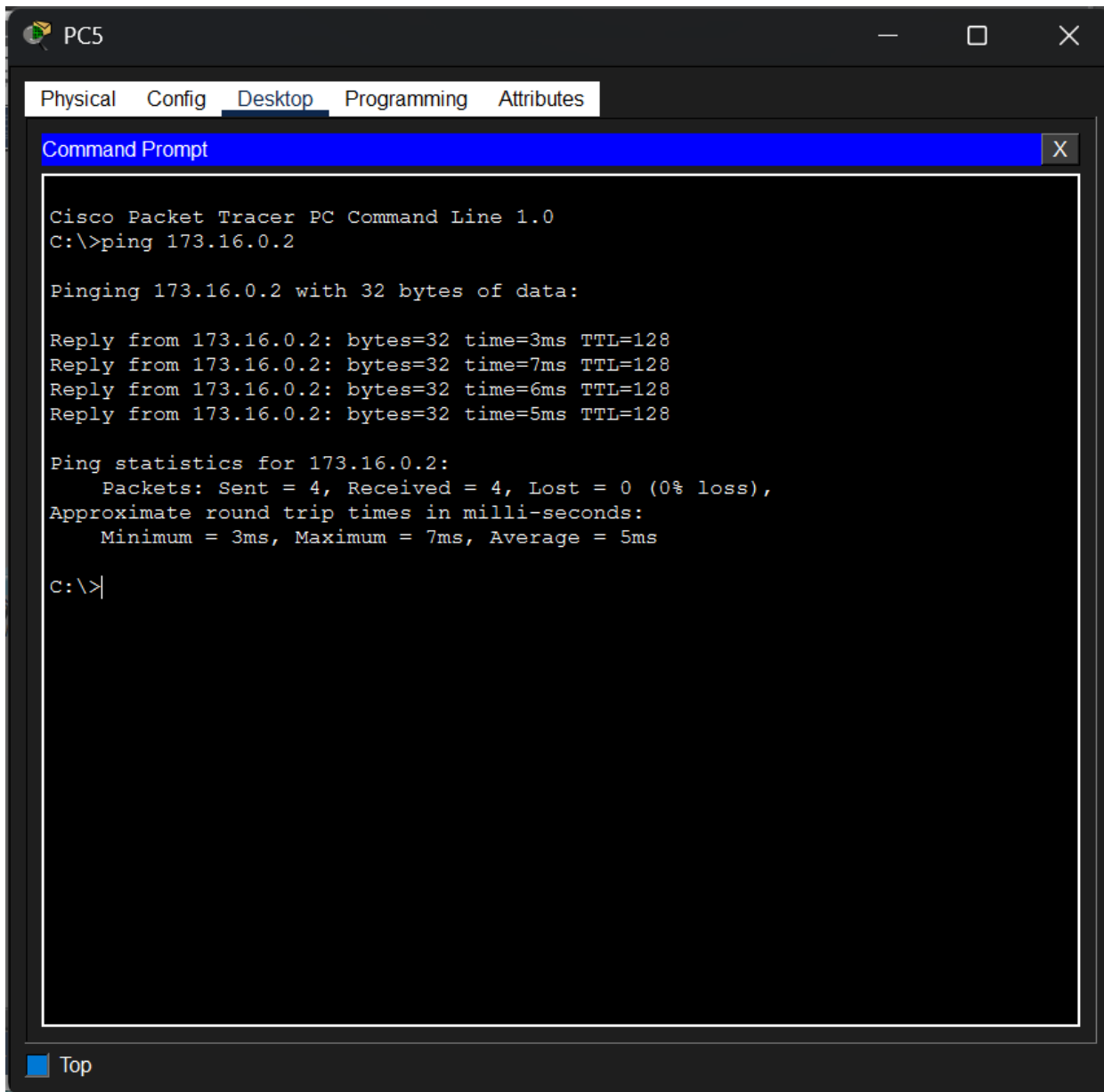
Pinging 172.16.0.2 with 32 bytes of data:

Reply from 172.16.0.2: bytes=32 time=14ms TTL=128
Reply from 172.16.0.2: bytes=32 time=7ms TTL=128
Reply from 172.16.0.2: bytes=32 time=6ms TTL=128
Reply from 172.16.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 6ms

C:\>
```

Top

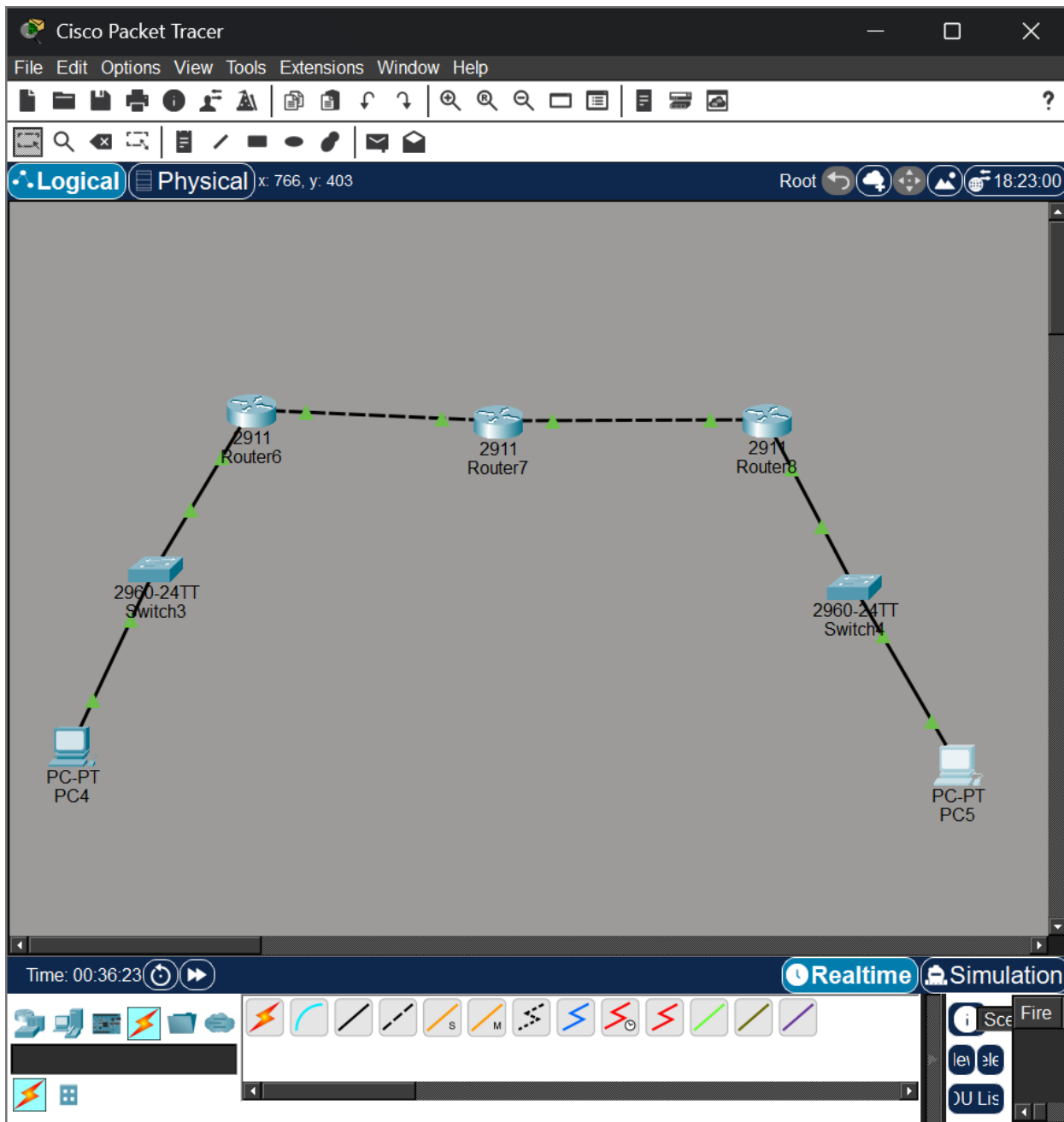


Brief Explanation on RIP Routing

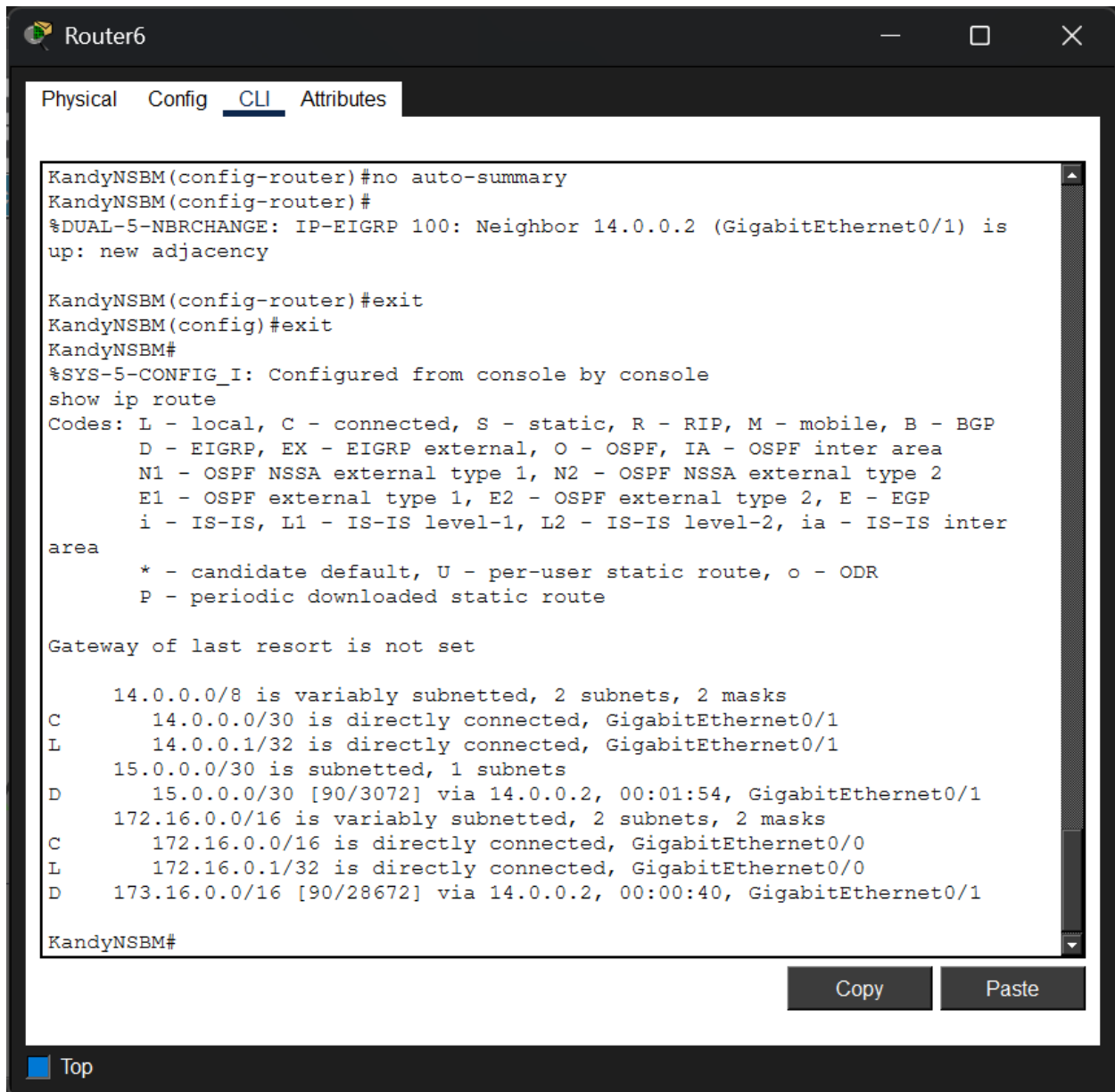
In this task, RIP (Routing Information Protocol) was configured on all routers to enable dynamic routing. RIP automatically exchanges routing information between routers and updates routing tables dynamically. The router rip, version 2, and network commands were used to advertise connected networks. Successful communication between all networks was verified using ping tests and routing table entries marked with R.

Part B – EIGRP Routing

1. Network diagram



2. Configuration commands and Screenshots of routing tables\



The screenshot shows a Cisco Router6 CLI window with the following content:

```
Router6
Physical Config CLI Attributes
KandyNSBM(config-router)#no auto-summary
KandyNSBM(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 14.0.0.2 (GigabitEthernet0/1) is
up: new adjacency

KandyNSBM(config-router)#exit
KandyNSBM(config)#exit
KandyNSBM#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/1
L       14.0.0.1/32 is directly connected, GigabitEthernet0/1
    15.0.0.0/30 is subnetted, 1 subnets
D       15.0.0.0/30 [90/3072] via 14.0.0.2, 00:01:54, GigabitEthernet0/1
    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       173.16.0.0/16 [90/28672] via 14.0.0.2, 00:00:40, GigabitEthernet0/1

KandyNSBM#
```

At the bottom right of the CLI window, there are two buttons: "Copy" and "Paste". At the bottom left, there is a "Top" link.

```
ColomboNSBM(config-router)#network 15.0.0.0
ColomboNSBM(config-router)#no auto-summary
ColomboNSBM(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 15.0.0.2 (GigabitEthernet0/1) is
up: new adjacency

ColomboNSBM(config-router)#exit
ColomboNSBM(config)#exit
ColomboNSBM#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    14.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       14.0.0.0/30 is directly connected, GigabitEthernet0/0
L       14.0.0.2/32 is directly connected, GigabitEthernet0/0
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       15.0.0.0/30 is directly connected, GigabitEthernet0/1
L       15.0.0.1/32 is directly connected, GigabitEthernet0/1
D       172.16.0.0/16 [90/28416] via 14.0.0.1, 00:02:53, GigabitEthernet0/0
D       173.16.0.0/16 [90/28416] via 15.0.0.2, 00:00:58, GigabitEthernet0/1

ColomboNSBM#
```

Copy

Paste

```
GalleNSBM(config-router)#router eigrp 100
GalleNSBM(config-router)#network 15.0.0.0
GalleNSBM(config-router)#network 173.16.0.0
GalleNSBM(config-router)#no auto-summary
GalleNSBM(config-router)#exit
GalleNSBM(config)#exit
GalleNSBM#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

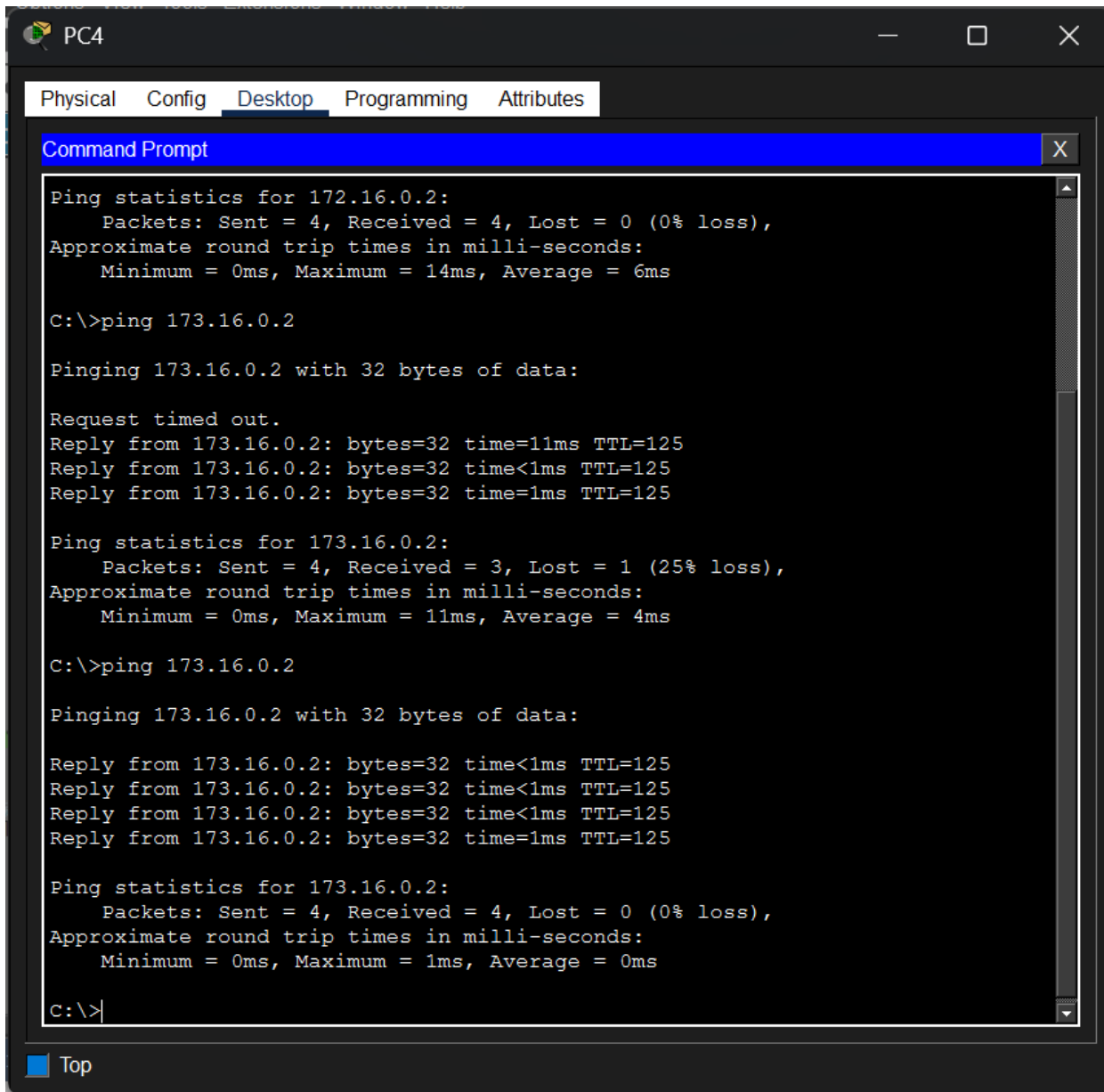
    14.0.0.0/30 is subnetted, 1 subnets
D       14.0.0.0/30 [90/3072] via 15.0.0.1, 00:01:44, GigabitEthernet0/1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       15.0.0.0/30 is directly connected, GigabitEthernet0/1
L       15.0.0.2/32 is directly connected, GigabitEthernet0/1
D       172.16.0.0/16 [90/28672] via 15.0.0.1, 00:01:44, GigabitEthernet0/1
    173.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       173.16.0.0/16 is directly connected, GigabitEthernet0/0
L       173.16.0.1/32 is directly connected, GigabitEthernet0/0

GalleNSBM#
```

Copy

Paste

5. Ping test results



The screenshot shows a window titled "PC4" with a dark theme. It has a tabbed interface with "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the results of two ping tests. The first test is for 172.16.0.2, which shows 0% loss and an average round trip time of 6ms. The second test is for 173.16.0.2, which shows a 25% loss and an average round trip time of 4ms. The Command Prompt window has a title bar with "X" and a scroll bar on the right.

```
PC4
Physical Config Desktop Programming Attributes
Command Prompt X
Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 6ms

C:\>ping 173.16.0.2

Pinging 173.16.0.2 with 32 bytes of data:

Request timed out.
Reply from 173.16.0.2: bytes=32 time=11ms TTL=125
Reply from 173.16.0.2: bytes=32 time<1ms TTL=125
Reply from 173.16.0.2: bytes=32 time=1ms TTL=125

Ping statistics for 173.16.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>ping 173.16.0.2

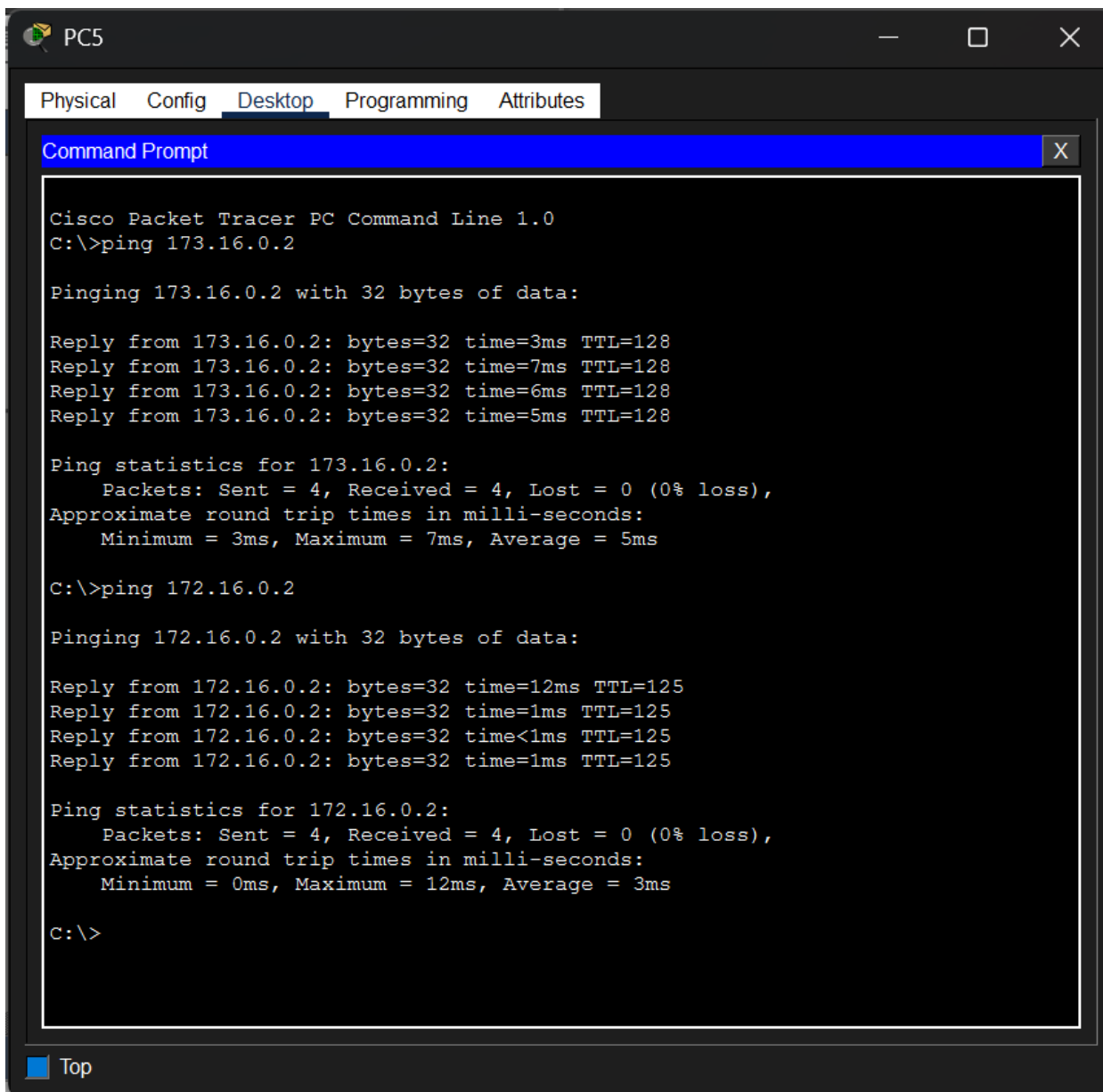
Pinging 173.16.0.2 with 32 bytes of data:

Reply from 173.16.0.2: bytes=32 time<1ms TTL=125
Reply from 173.16.0.2: bytes=32 time<1ms TTL=125
Reply from 173.16.0.2: bytes=32 time<1ms TTL=125
Reply from 173.16.0.2: bytes=32 time=1ms TTL=125

Ping statistics for 173.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Top



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 173.16.0.2

Pinging 173.16.0.2 with 32 bytes of data:

Reply from 173.16.0.2: bytes=32 time=3ms TTL=128
Reply from 173.16.0.2: bytes=32 time=7ms TTL=128
Reply from 173.16.0.2: bytes=32 time=6ms TTL=128
Reply from 173.16.0.2: bytes=32 time=5ms TTL=128

Ping statistics for 173.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 7ms, Average = 5ms

C:\>ping 172.16.0.2

Pinging 172.16.0.2 with 32 bytes of data:

Reply from 172.16.0.2: bytes=32 time=12ms TTL=125
Reply from 172.16.0.2: bytes=32 time=1ms TTL=125
Reply from 172.16.0.2: bytes=32 time<1ms TTL=125
Reply from 172.16.0.2: bytes=32 time=1ms TTL=125

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>
```

Brief Explanation on EIGRP Routing

In this task, the RIP configuration was removed and EIGRP (Enhanced Interior Gateway Routing Protocol) was configured using Autonomous System number 100. EIGRP dynamically shares routing information and provides faster convergence and better performance compared to RIP. The router eigrp 100 and network commands were used to advertise networks. Successful communication between all networks was verified using ping tests and routing table entries marked with D.

Conclusion

This assignment successfully demonstrated the configuration and implementation of router networking concepts using Cisco Packet Tracer. Basic router configuration was performed by setting hostnames, console passwords, and login authentication to improve device security and management. Static routing was configured to establish communication between multiple networks using manually defined routes. In addition, dynamic routing protocols RIP and EIGRP were implemented to automatically exchange routing information between routers. Connectivity between all networks was successfully verified through ping tests and routing table analysis. Overall, this assignment improved practical knowledge of router configuration, IP addressing, static routing, and dynamic routing protocols in computer networks.

GitHub Link

<https://github.com/pdchamodiwya/CNRoutingAssignment1>